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SUMMARY

Robotics & Mechanical Engineer with experience in robotics simulation, reinforcement learning, and product design. Strong track record across robot learning, humanoid control, and mechanical systems design. Published in Nature, awarded a U.S. patent, and co-PI on \$1.3M funded project. Skilled in PyTorch, RL algorithms, ROS2, IsaacSim, and CAD/FEA.

CORE SKILLS

- Robotics & AI: Reinforcement Learning (PPO, SAC, TD-MPC), Multi-agent systems, Motion Imitation, Humanoid- Locomotion, RL Reward Design, ROS2, IsaacSim, Embedded systems, Arduino
- Programming: Python, PyTorch, Stable Baselines, Pandas, Git, Data Visualization
- Mechanical Design: SolidWorks, Ansys, CAD/CAE, GD&T, ASME/EN standards
- Other: Product Development, Life Cycle Analysis, Multimodal AI, Vision-Language Models

EDUCATION

M.S. Mechanical Engineering (Robotics Focus), Northwestern University, Evanston, IL | 2019 – 2021

- Thesis: Soft Robotics Design & Simulation
- Coursework: Reinforcement Learning, Control Systems, Robot Manipulation, Machine Dynamics

PUBLICATIONS & PATENTS

- Nature Scientific Reports (2022): KnitMorphs – Numerical Simulation of Knit-Based Morphable Structures
- U.S. Patent: Systems & Methods for Dieless Composite Forming (approved)

FUNDING & CERTIFICATIONS

- Co-PI on \$1.3M DOE project for high-temperature particle in-line mass flow sensor
- NVIDIA GTC 2025 – Building AI Agents with Multimodal Models (certified)

PROJECTS IN ROBOTICS (chronological)

- ACTION100M: Implementation of the Action100M data pipeline for generating hierarchical video action annotations: Code: <https://github.com/skr3178/ACTION100M>
- Humanoid-twin-world-model A digital twin of Unitree G1 humanoid in a 3d Gaussian Splatting scene. The humanoid is controlled by arrows. It used world model API/ sample cobble path files: <https://github.com/skr3178/humanoid-twin-world-model>

- TopoDiffuser: Diffusion based framework for multimodal trajectory prediction. Recreated the code and implementation of the work on open source from scratch- Lidar only prediction reaches accuracy within 1 meter: <https://github.com/skr3178/TopoDiffuser>
- CoPilot4D (in progress): Recreated the code and implementation of the work on open source from scratch- Lidar prediction 3s into future: <https://github.com/skr3178/copilot4D>
- DreamerV4 (in progress-limited by compute): Recreated the code and implementation of the work on open source from scratch on a much compact dataset (mineRL) than used in paper: <https://github.com/skr3178/DreamerV4>
- Humanoid World model: Recreated the code and implementation of the work on open source from scratch: https://github.com/skr3178/Humanoid_world_model
- Genie Robot World model: implementation of Genie: Genie: Generative Interactive Environments from scratch. Implementation tested on PONG dataset: https://github.com/skr3178/Genie_google
- Pi0 policy: Developed deep understanding of the SOTA VLA model Pi0- flow matching, action chunking, action embeddings to existing model: <https://github.com/skr3178/openpi>
- Eureka on Piano playing dextrous hand: Repurposed the LLM based reward formulation pipeline for piano playing by dextrous hand robot: https://github.com/skr3178/Eureka/tree/mujoco_piano
- Eureka: Reviewed code for generating reward design using LLM, comparing hand and Humanoid env reward from LLM with isaacenv results: <https://github.com/skr3178/Eureka>
- Linear Kalman filter and sensor fusion: Completed Course on Coursera developing deep mathematical intuition behind sensor fusion and use of Kalman filter for robotics: https://github.com/skr3178/kalman_filter_notes
- Motion Diffusion model: Reviewed code for generating motion control of skeleton frames. My understanding is here: <https://github.com/skr3178/motion-diffusion-model>
- Motion retargeting from Miku Miku Dance to Unitree Humanoid: <https://github.com/skr3178/MikuMotionTools>
- iOS app for hand-tracking: Made an app to detect hand pose estimation from photo, youtube videos or live camera stream, model compression using onnx <https://github.com/skr3178/hand-landmarker-ios>
- ROS 2 Nav2 [Navigation 2 Stack] - with SLAM and Navigation: Understand the Nav2 Stack with ROS2 - SLAM, Mapping, Navigation, Gazebo Simulation, Python Code - Step by Step [Certificate](#)
- DeepMimic: Analysed and uncovered the code behind imitation learning of robots from motion capture data using AMP algorithm. GitHub: https://github.com/skr3178/AMP_MUL
- Vision Language Action (VLA) Model – Decoder-only Transformer: Created a vision language action model predicting actions for the next 20 steps using PushT env. GitHub: https://github.com/skr3178/Dot_policy
- Unitree Humanoid Robot: Reinforcement learning reward design for gait and running – trained on IsaacSim.
- Human in the Loop (HIL): Training with RL coupled with human intervention for a pick-and-place robot (keyboard input). GitHub: <https://github.com/skr3178/gym-hil>
- S0101 + IsaacSim: Creating pick-and-place robot env and training with RL and humanoid Gait: GitHub: <https://github.com/skr3178/S0101-IsaacSim>

- Deep Reinforcement Learning: Explored algorithms: Q-learning, Value & Model-based methods. Designed reward functions for cooperative-competitive agents, and implemented Hidden Markov agents. GitHub: https://github.com/skr3178/Dot_policy
- Probabilistic Models: Learnt background mathematics, Bayesian Learning, Expectation Maximization, Value function, Entropy, Directed & Undirected Graphs.
- TD-MPC (Temporal Difference Model Predictive Control): Broke down and explained TD-MPC algorithm. GitHub: https://github.com/skr3178/tdmpc_explained
- Vision-Language & HUMANOID-BENCH Experiments: Experimentation with vision-language model combinations for humanoid simulations.
- Wheeled Robot Manipulation: Representations of configuration & spatial motion of rigid bodies/robots using screw theory. Forward, inverse, differential kinematics, robot dynamics, trajectory planning, motion control, mobile manipulation. GitHub: https://github.com/Sangram-Rout/ME449-Robotic_manipulation
- Dynamics of Robotic Systems: Machine Dynamics course: variational analysis applied to simulation & mechanism design. Simulated triple pendulum in Python. (GitHub available)
- ROS2 Course (Independent): Nodes, publisher, subscriber, navigation with TurtleBot, URDF of wheeled robot.
- Multi-Agent Reinforcement Learning (Stable Baselines): Studied and applied algorithms (DQN, PPO, SAC) to games (Atari, Ant, Humanoid). GitHub: https://github.com/skr3178/reinforcement_learning

EXPERIENCE

Mechanical Engineer

VSParticle, Delft, Netherlands | Aug 2024 – Apr 2025

- Co-Led development of a multi-material Nano-printer for novel material synthesis.
- Reimplemented gas recycling system using microcontrollers, sensors & valves.
- Designed custom Faraday enclosures & PCB design for substrate holders for electrocatalysis applications.

Research Assistant – Human Computer Interaction

Northwestern University, Evanston, IL | Jul 2023 – Jan 2024

- Prototyped mechanical logic gates (AND, XOR, NOR) using nitinol actuators.
- Demonstrated half/full adder circuits with 3D-printed parts & copper-plated contacts.

Mechanical Engineer

Heliogen, Pasadena, CA | Aug 2021 – Apr 2023

- Improvement of design and production of multi-ton 2m solar receiver with CAD/FEA (thermal, stress, drawings).
- Conducted Life Cycle Analysis for 26.6 MW solar plant, providing overall CO₂ reduction strategies.
- Delivered wind data analysis & visualization of 5 Million datapoints to optimize receiver particle loss.
- Design of a novel particle valve.

Mechanical Design Engineer

EagleBurgmann Expansion Joints, Denmark | Oct 2014 – Aug 2019

- Designed large-scale expansion joints under 800°C & 200 bar.
- Applied ASME & EN standards for high-pressure components.
- Improved designs with \$2M annual cost impact.

Powertrain Engineer (Formula SAE)

Ashwa Racing, Bangalore | Jan 2011 – Apr 2014

- Designed intake/exhaust for 600cc engine → 40% more power & torque.